3 Results and Discussion

Cloudra is designed around two transitions in reaction-mechanism research: translating chemical intent into concrete geometric operations, and assembling local computational results into support for a scientific conclusion. Using autonomous research trajectories, paired local comparisons, and interventions on internal model states, this section examines how information contributes to these transitions and connects molecular representations, geometric operators, the evidence DAG, and diagnostic feedback.

3.1 Experimental systems and task coverage

The 50 chemical problems in CloudraBench cover elementary transformations, multistep processes, competing pathways, stereochemistry, and electronic-state-dependent judgments. They comprise 20 Easy, 10 Normal, 10 Hard, and 10 Impossible tasks. Each task combines a chemical category with a research requirement. For example, an electrocyclic reaction may require identifying a reaction pathway or comparing the stereochemical outcomes of different directions of rotation. The same representations of reactants and products may also describe rearrangements between different configurations or questions of accessibility on a specified electronic surface. In a semantic analysis of task names, Diels–Alder reactions and cyclobutene ring openings each formed neighborhoods spanning difficulty levels, reflecting the intersecting distributions of chemical topics and research requirements.

The process analysis uses 48 fully archived autonomous runs: four system configurations—Guided, Record, Standard, and Brief-errors—applied to four representative tasks, with three repeats per condition. The tasks were fluoride-mediated elimination of chloroethane (E1), thermal decarboxylation of malonic acid (NB1), a stereochemical problem involving 3-methylcyclobutene ring opening (HA1), and a cyclobutadiene self-rearrangement problem involving rectangular and square configurations (IA1). These tasks respectively require preorganization of a reaction center, continuity between successive steps, comparison of competing configurations, and assessment of a specified potential-energy-surface description. The formal runs proceeded without human intervention. Archived conversations, tool calls, visible outputs, and computational artifacts provide the sources for the analysis. A total of 6,087 relevant scientific tool-result records were extracted to trace relationships between research objectives, program responses, and subsequent actions.

The archived runs used DeepSeek-v4-flash. Local diagnostics used Qwen3.8:27b with paired inputs constructed around source structures or evidence chains. These diagnostics varied molecular representation, option order, geometric information, evidence references, or reasoning mode, and recorded actual choices and generated token counts. The original trajectories show how problems arose during research; paired inputs reveal how specific information contributed to judgments; and internal-state interventions further trace the influence of measured facts on selection. The following analysis develops the two design themes through these three levels of evidence.

3.2 From chemical intent to reliable geometric operations

3.2.1 Molecular identity provides a persistent reference for operations

A geometric modification first requires identifying the molecule, atoms, and reaction center to which it applies. A molecule can be represented by different SMILES traversals, while a molecular formula can describe different connectivities. To examine how the model handles these distinctions, we constructed 16 molecular-identity pairs covering identical identities, connectivity changes within the same molecular formula, changes in connectivity and charge location, double-bond stereochemistry, and protonation changes. Each pair was presented using the original strings or equivalent rewritings and with two answer orders, yielding 64 judgments.

The model answered 58 correctly. Equivalent rewriting changed the semantic judgment in 4 of 32 matched representation comparisons; reversing answer order changed the judgment in 6 of 32 matched option-order comparisons. All six of the latter cases involved repeated selection of the letter A. One C_5H_8 pair in Figure 5 illustrates this behavior directly: an endocyclic and an exocyclic double bond define different connectivities, yet the model answered A under all four conditions, although A meant “same” in one order and “different” in the other. This result locates representation sensitivity in judgments about the chemical objects themselves and indicates the need for stable identity references throughout a research process.

Cloudra assigns this role to molecular topology, atom mapping, and reaction centers. These references allow subsequent measurements, fragment movements, and endpoint comparisons to remain attached to the originally specified objects as the model moves between descriptions and configurations. Atom correspondence also connects product-endpoint alignment, path interpolation, and reaction-center monitoring, allowing local geometric operations to proceed with a consistent set of chemical identities.

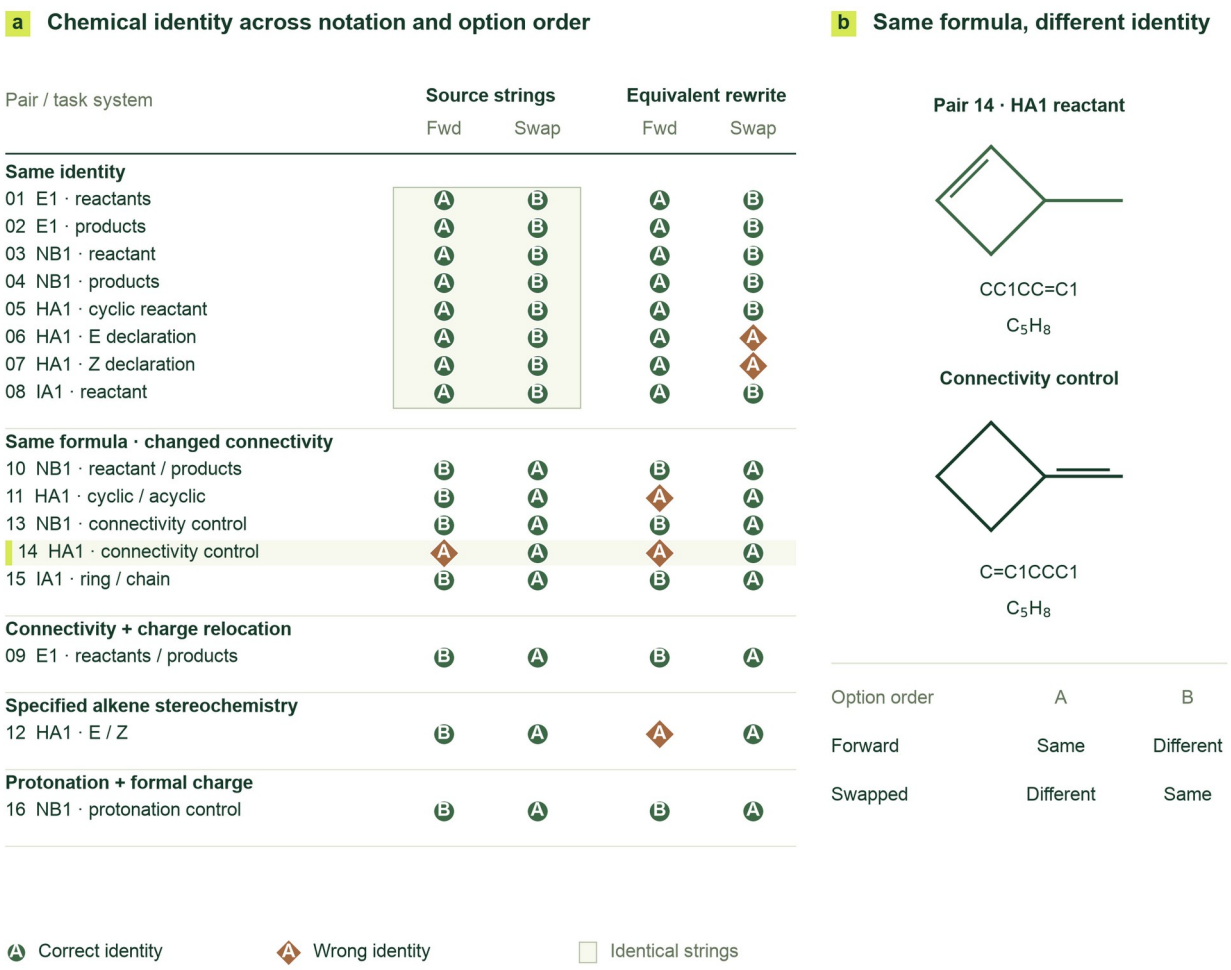


Figure 5. Molecular-identity judgments under changes in molecular representation and option order. Each of the 16 molecular pairs has four input conditions. Circles and diamonds indicate correct and incorrect judgments, respectively; the letters retain the model's actual answers. Shaded regions denote identical-string conditions. The right panel shows the two C_5H_8 connectivities in pair 14 and the meanings of the answer options.

3.2.2 Geometric measurements reduce repeated numerical reasoning

After identifying the research objects, the model must extract spatial relationships relevant to an operation from their coordinates. We selected four actual structures from the NB1 trajectories and asked the model to compare two specified interatomic distances. Each structure was presented with both answer orders and with either a complete XYZ geometry or program-measured distances. These two information conditions were further crossed with direct-answer and thinking modes, producing 32 native chat calls (Figure 6).

In direct-answer mode, all eight responses to XYZ inputs selected A, with four correct answers. Providing measured distances produced eight correct answers and preserved the semantic answer when option order was reversed. With thinking enabled, the XYZ condition produced seven correct answers; the remaining response reached the output limit of 1,024 generated tokens. Increasing the budget for that input alone yielded a correct answer within 1,353 generated tokens. All eight answers in the measured-distance condition were correct. Across the eight paired comparisons with thinking enabled, the median generated token counts were 807.5 for XYZ inputs and 86 for measured distances.

These results suggest that geometric judgment combines atom identification, numerical calculation, and relational comparison. Additional reasoning can perform these local calculations, while program measurements directly supply the relational quantities needed for the subsequent decision. Cloudra therefore allows the model to express objectives through bond lengths, bond angles, dihedral angles, and fragment relationships, and uses deterministic operators to change the coordinates. This division of work assigns repeated numerical operations to the program and makes the resulting changes directly inspectable, consistent with the allocation of responsibilities in program-aided language models.[15] Descriptions of steric crowding, reactant preorganization, and local fragment rotation build on this foundation to use spatial relationships in selecting and constructing candidate configurations.

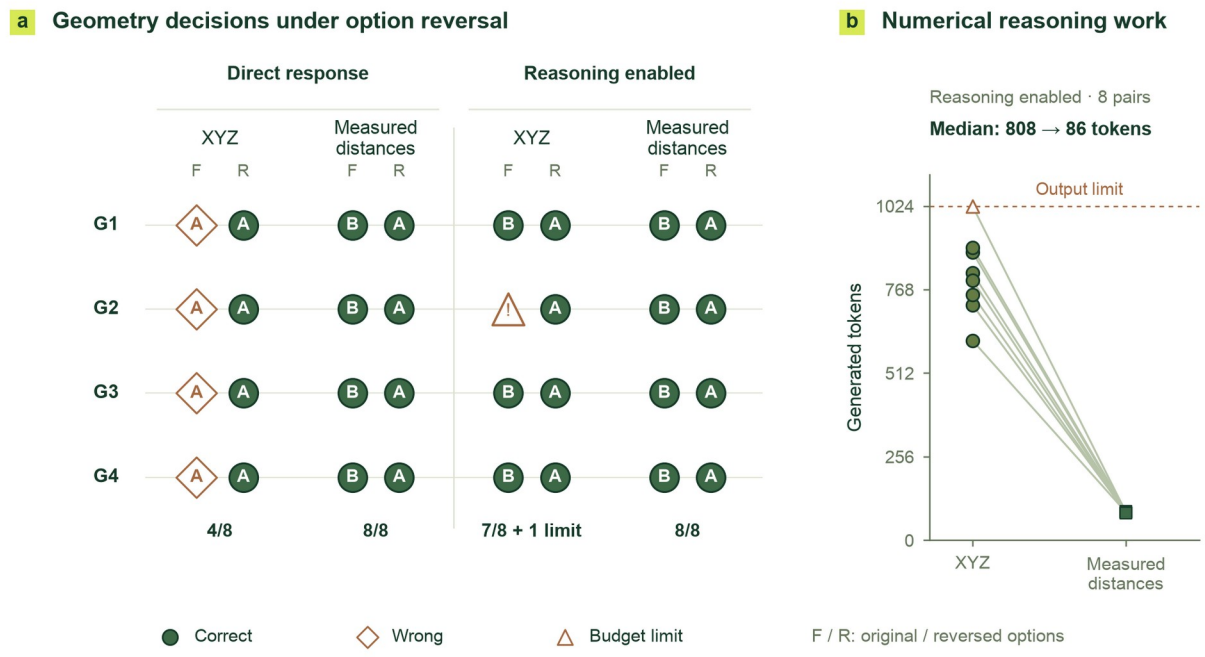


Figure 6. Paired comparisons of coordinate inputs, program measurements, and reasoning modes. The left panel shows responses for four source structures under four conditions and two option orders; triangles indicate responses that reached the output budget. Each line in the right panel connects the actual generated token counts for the two information conditions for the same structure and option order with thinking enabled.

3.2.3 Feedback must match the specified atoms and geometric objective

Deterministic execution must remain aligned with chemical intent. An actual E1 trajectory provides a complete example. The model sought to bring the dihedral angle defined by the transferring hydrogen H7, C2, C3, and leaving atom Cl4 close to 180°. It rotated the fragment on the C3 side about the C2–C3 axis by +120° and −120°, producing candidates A and B, and set the C3–Cl4 distance to 2.35 Å. The representative dihedral in the default structure report involved H5, whereas the current objective involved H7. The model then explicitly measured H7–C2–C3–Cl4 in both candidates, obtaining approximately 59.4° and 179.4°, and used candidate B in its next actual submission.

Figure 7 shows Newman projections along C2→C3 drawn directly from the saved candidate coordinates. Candidate B places the H7 target angle within 0.6° of 180°, compared with a residual of 120.6° for A. The corresponding residuals for the H5 comparison were 124.2° and 115.8° in A and B, respectively. These measurements reveal the object dependence of feedback: a reading acquires its chemical meaning through its correspondence to the currently specified atoms and operational objective.

Local comparisons using the same measurements further connect information to action. When only H5 measurements were provided, the model requested an additional H7 measurement under all three cyclic option permutations. Adding H7 measurements led it to select B under all three permutations. Two further paired tests of information sufficiency also shifted from requesting additional measurements to judging the information sufficient when the target quantity was added. Atom mapping, internal coordinates, fragment operations, and feedback can therefore be understood as one sequence of research actions: identify the target object, make the geometric change, measure the target quantity, and select the next candidate. Preorganization, rigid-body alignment, and path preparation use the same correspondence to carry reaction intent into subsequent exploration of the potential energy surface.

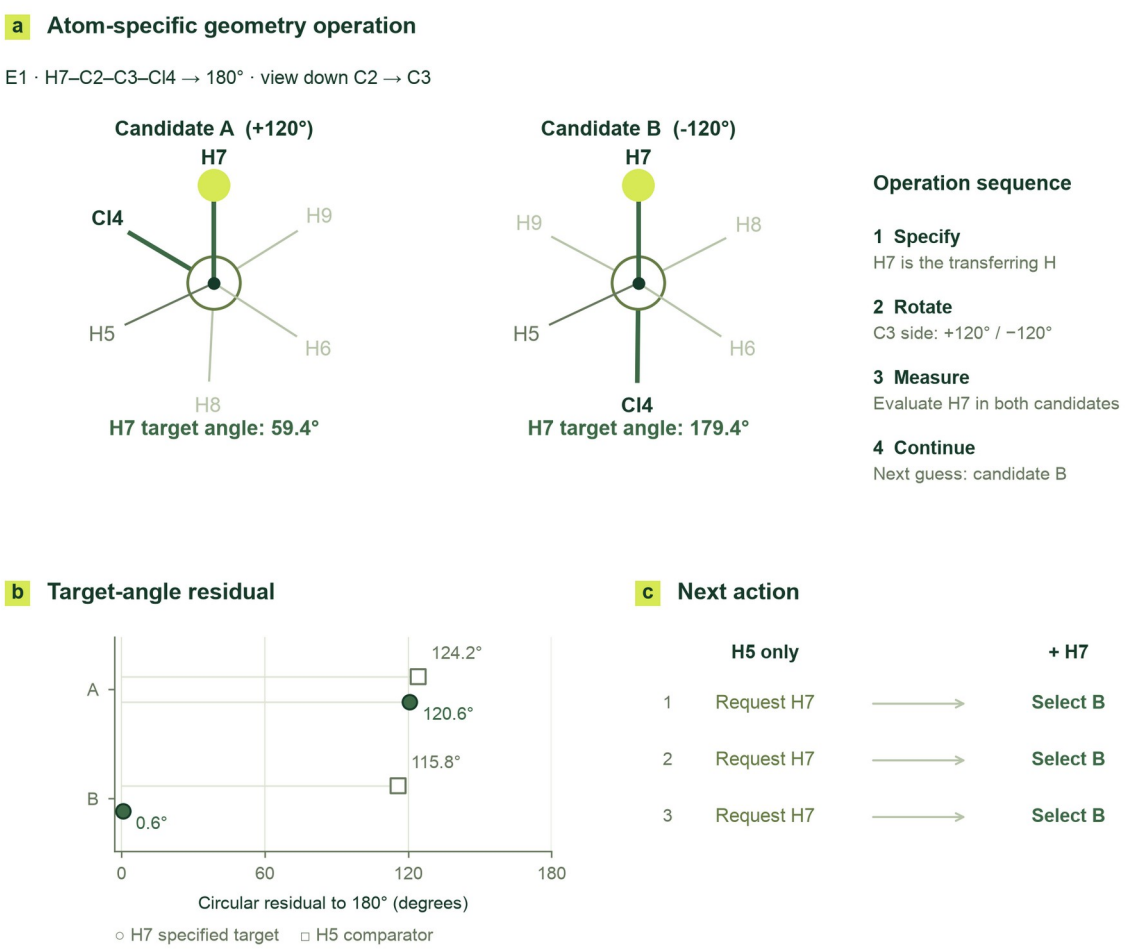


Figure 7. Alignment with the target and candidate selection in E1. Newman projections of candidates A and B are drawn using the projection direction of the original coordinates. H7 is the specified target atom. The lower panel compares the circular residuals of H7 and H5 relative to the target angle and shows the change from requesting a target measurement to selecting B across three local option permutations.

3.2.4 Internal transfer of measured facts affects selection

To examine how the model uses supplied measurements, we performed activation replacement at a single token position in the same local quantized model.[16] One distance in the input was changed from 2.029 Å to 1.029 Å, while the other remained at 1.248 Å, reversing the correct comparison answer. We then transferred the residual-stream state at the numeric token position from the original input into the modified input at layers 7, 15, 31, and 47, and read the full-vocabulary probability assigned to the answer supported by the original fact.

The effect varied with layer and option order (Figure 8). With reversed options, the probability of A, the original-fact answer, increased from 0.074 for the modified input to 0.956 and 0.718 after replacement at layers 7 and 15, respectively. Replacements at layers 31 and 47 yielded 0.072 and 0.070. With the original option order, the probability of answer B changed from 0.009 to 0.029 and 0.015 after replacement at the first two tested layers, while the last two remained close to the modified-input baseline. Self-replacement controls and control replacements at an unchanged prefix position produced zero logit differences across the full vocabulary.

This internal intervention extends the analysis from supplying measurements to examining how measurement information contributes to selection. States at a specified numeric position can change subsequent answer preferences, with the effect depending on both the transfer location and the organization of the answers. Together with the behavioral option-order comparisons, this finding suggests that tool outputs should maintain correspondence among facts, objects, and the decision being made. Cloudra's structured measurements thus serve as both the results of geometric calculations and inputs to the model's next research judgment.

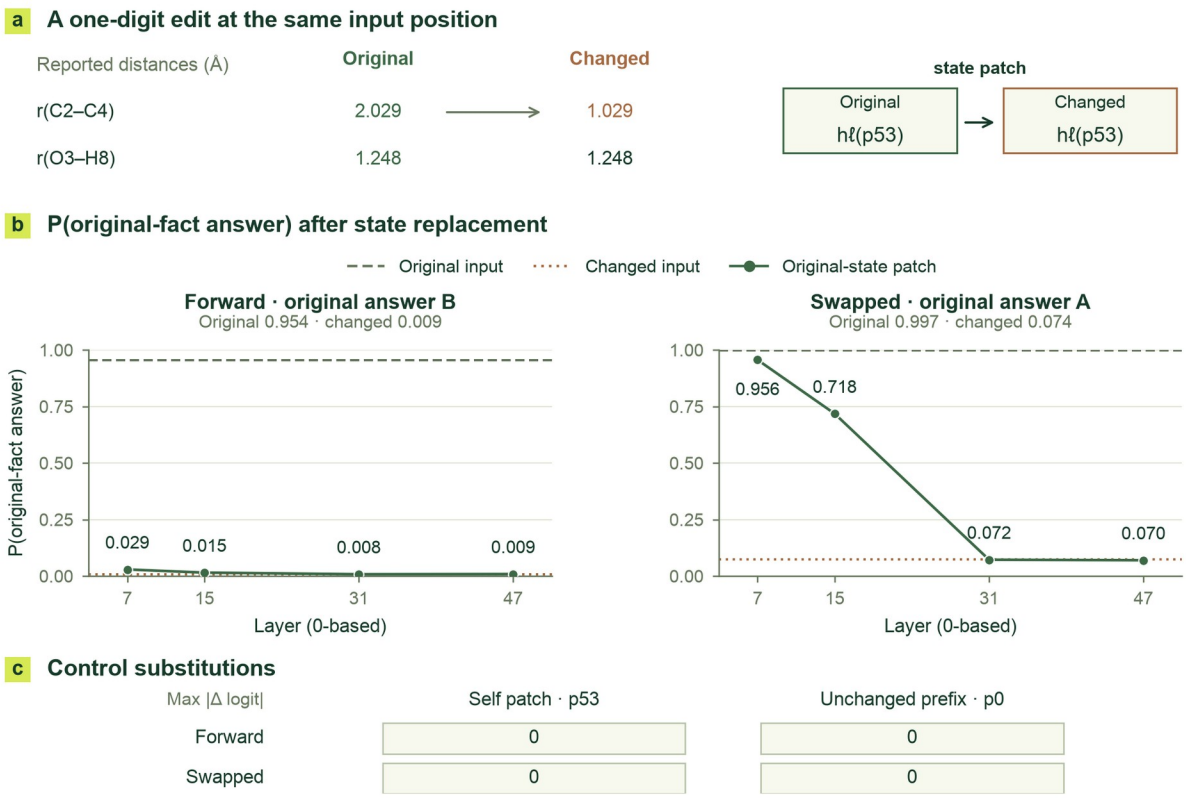


Figure 8. Internal-state replacement for a single measured fact. The upper panel shows the changed distance value and residual-stream replacement. The middle panels plot the probability of the original-fact answer and the two input baselines under each option order. The lower panel shows self-replacement and prefix-replacement controls. Layer and token indices are zero-based.

3.3 From local computational results to evidence-supported scientific delivery

3.3.1 The DAG preserves scientific support relationships among results

Once geometric objects enter computation, optimized structures, energies, frequencies, and paths must be assembled into support for the same reaction step. The local validity and attribution of each result jointly determine this support chain: frequencies must correspond to the declared saddle point, the IRC must start from that saddle point, and optimizations and frequencies for the final frames on both sides must correspond to the respective endpoints. The DAG records these relationships on species and elementary steps, providing a persistent research index as results accumulate. Execution state, numerical convergence, and scientific support are recorded separately, allowing subsequent work to wait for a calculation, revise and rerun it, or continue assembling evidence.

We extracted four source evidence chains from the archive and constructed versions that preserved the links or changed a single reference. Crossing these with flat and linked representations and two answer orders yielded 32 inputs. The altered references concerned the frequency geometry, IRC starting geometry, endpoint-optimization input, or target species; local termination, convergence, energy, and frequency values were held constant. Figure 9a illustrates the E1 case: changing the frequency geometry reference from the saddle point to an existing endpoint changes how that frequency result supports the current step.

All 16 judgments with linked representations were correct, compared with 14 of 16 with flat representations. Both errors occurred when a reference had been changed and the options reversed, in E1 and the second NB1 step. In subsequent native follow-up calls on two selected source chains, enabling thinking yielded correct judgments for all 16 judgments of the evidence relationships, including the two previously incorrect cases. This result suggests that explicit links reduce the model's work in locating and assembling computational relationships, consistent with observations that graph encoding affects relational reasoning in language models.[17] In Cloudra, this representation is further bound to actual computational artifacts, allowing scientific facts to accumulate continuously for the same species, step, and attempt.

3.3.2 Diagnostics turn evidence gaps into concrete research actions

Linkage states must enter feedback accessible to the model to support correction. In one NB1 case, the complete response already contained the blocking atom pair, but the model initially printed only the first 2,500 characters of the full summary, leaving that object outside the visible output. A targeted read of the blocking field then exposed the relevant information. The

model revised the reaction center within the same attempt, and a subsequent optimization job entered the queue. This trajectory shows that output selection and object localization lie between fact generation and the next action.

Two other trajectories illustrate the interaction between scientific guidance and operational interfaces. In a Brief-errors HA1 case, the experimental configuration removed the specific diagnosis of an IRC-endpoint linkage problem. The model then changed only the source-minimum reference, and resubmission retained the linkage gap. In a Guided NB1 case, the underlying check found zero imaginary frequencies, while the Computing Agent saw a mode-error category. The Coordinator then supplied a specific retry plan, which the Computing Agent used to prepare a new guess and revised saddle-point optimization settings, incorporating the changes into the computational definition actually executed. Together, these cases show that recovery requires connecting the problematic evidence to the object or calculation that needs revision. Research on model self-correction through external tool feedback likewise emphasizes this connection.[18]

This explains how Cloudra organizes scientific diagnostics and multi-agent collaboration. Computational agents revise specific attempts using structural and path feedback; coordinating agents use the results to continue a route or switch candidates; and supervising agents review the available support against task requirements. Asynchronous scheduling and persistent execution retain ongoing calculations, allowing each role to resume work on the same objects when results return. Shared objects and evidence connect the judgments of different roles, while subsequent calculations supply new facts for the next decision.

3.3.3 Outstanding requirements must remain part of completion decisions

Scientific delivery also requires a global judgment: whether the available results satisfy the research requirements. In an actual IA1 trajectory, the DAG report explicitly retained unmet evidence requirements and an empty waiver list, while a Supervisor had already endorsed part of the computational evidence. The model nevertheless attempted to end the research as complete. The program again returned a decision to continue and placed the missing short_irc requirement back into context; the model immediately changed its ending request to partial completion (Figure 9c). This behavior indicates that local results, review comments, and global research requirements may all contribute to the model's completion decision. Persistent scientific state provides a common reference for them.

The model can also use explicit state directly. In 12 comparisons of authoritative sources constructed around this case, every judgment followed the existing records of outstanding, unwaived requirements. When supplied either the latest report alone or a history window containing 12 original messages, all 12 responses across three state questions and two answer orders likewise retained the unmet requirements. The DAG can therefore provide an explicit starting point for the next round of reasoning: which objects the available support belongs to, what gaps remain, and whether to obtain further evidence or submit partial results that reflect the current state. Supervisor review helps interpret routes, methods, and the scope of exceptions.

The two design themes converge here. On the geometric side, Cloudra preserves correspondence between chemical intent and atoms, fragments, and target quantities. On the evidence side, it preserves correspondence between scientific claims and computational results, support relationships, and outstanding requirements. Internal coordinates allow the model to advance through executable geometric changes; the DAG allows the resulting calculations to contribute to a cumulative scientific judgment. These designs serve a common research process in which model-proposed objectives remain connected to objects that can be manipulated, changes that can be observed, and evidence that can support delivery.

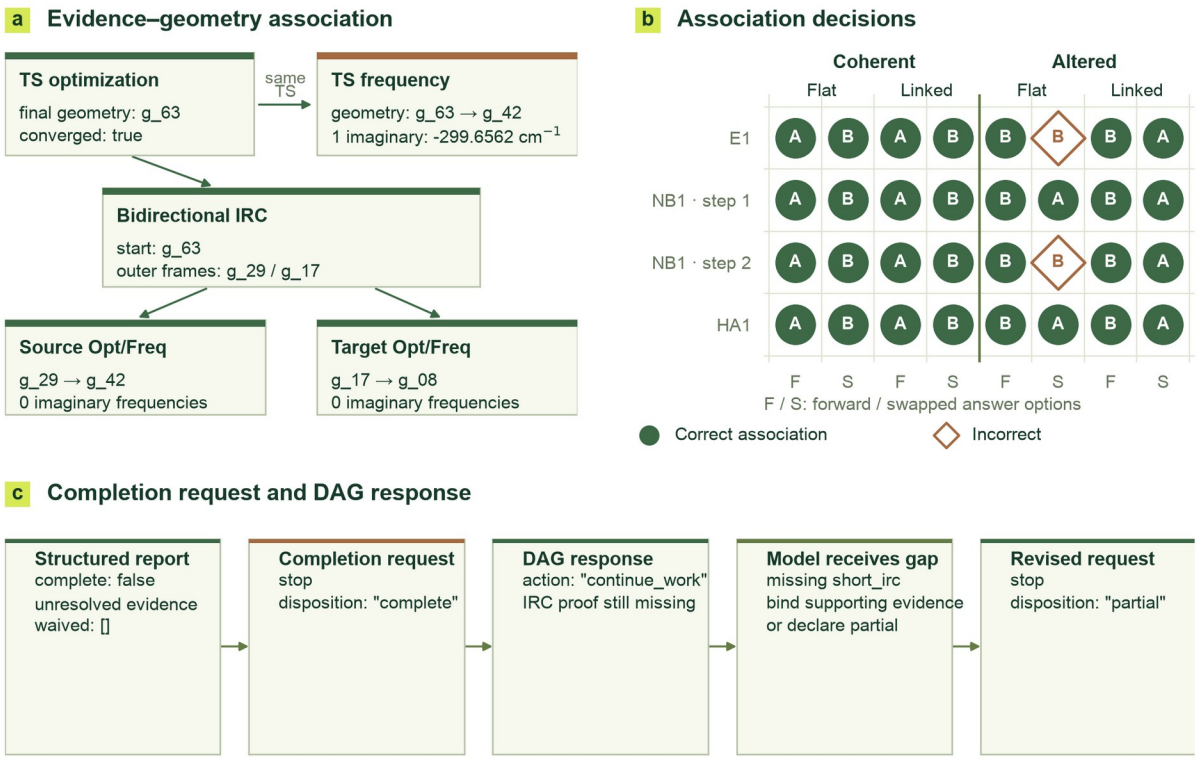


Figure 9. Evidence links and completion decisions. a, A changed frequency-geometry reference in the E1 source chain. b, The 32 responses for four source chains under crossed conditions of linkage status, representation, and option order. c, The behavioral sequence in the original IA1 trajectory, from the report of outstanding requirements through a completion request and DAG feedback to a partial-completion request.

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